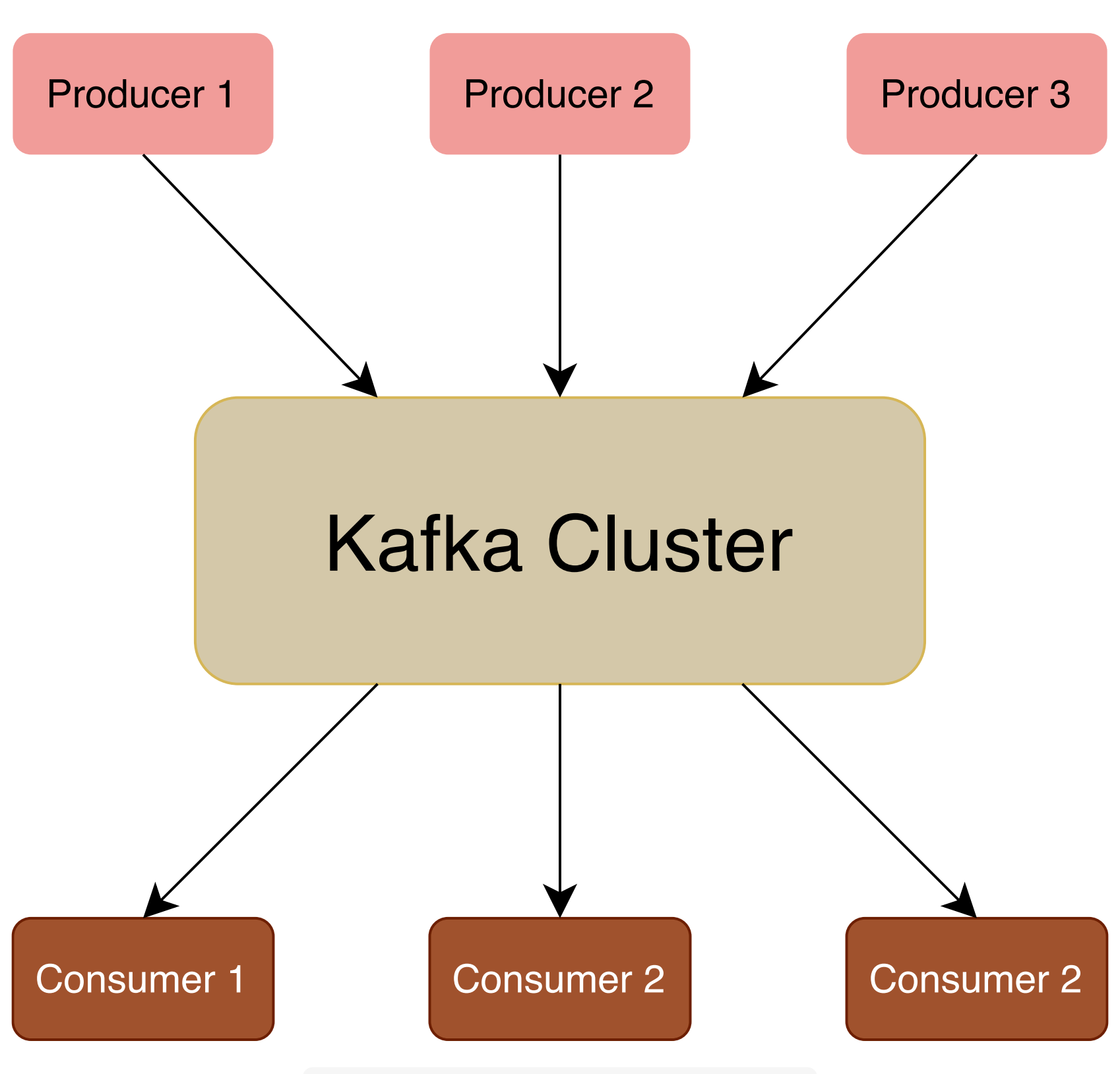
Introduction to Kafka

What is Kafka?

Apache Kafka is an open-source **publish-subscribe**-based messaging system. It is **distributed**, **durable**, **fault-tolerant**, and **highly scalable** by design. Fundamentally, it is a system that takes streams of messages from applications known as producers, stores them reliably on a central cluster (containing a set of brokers), and allows those messages to be received by applications (known as consumers) that process the messages.



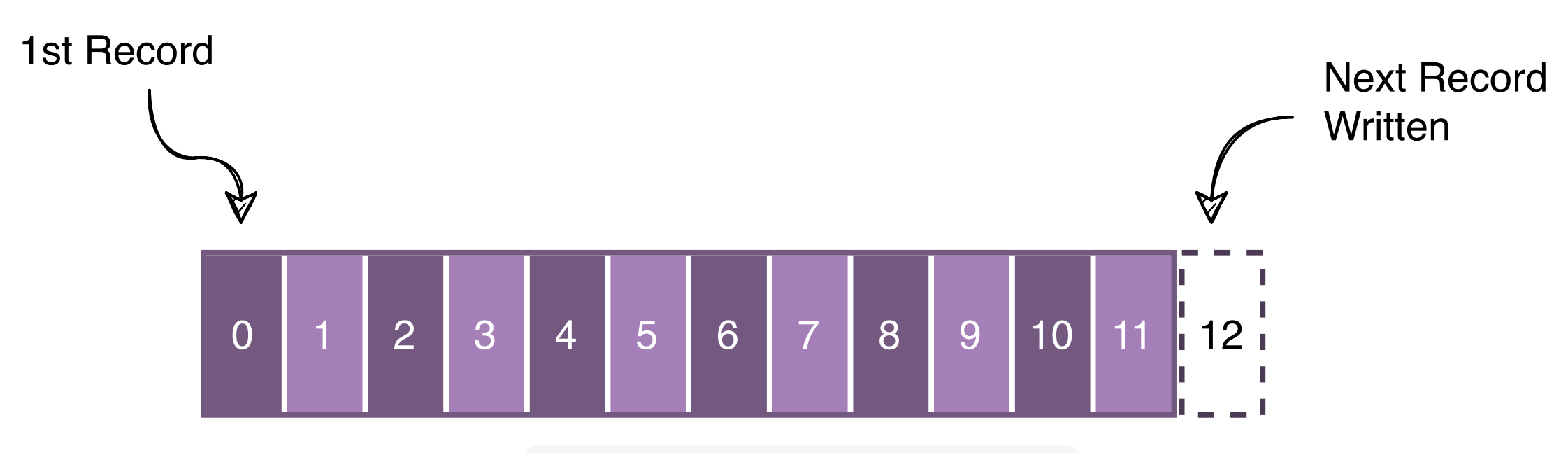
A high-level view of Kafka

Background

Kafka was created at LinkedIn around 2010 to track various events, such as page views, messages from the messaging system, and logs from various services. Later, it was made open-source and developed into a comprehensive system which is used for:

1. Reliably storing a huge amount of data.
2. Enabling high throughput of message transfer between different entities.
3. Streaming real-time data.

At a high level, we can call Kafka a distributed **Commit Log**. A Commit Log (*also known as a Write-Ahead log or a Transactions log*) is an **append-only** data structure that can persistently store a sequence of records. Records are always appended to the end of the log, and once added, records cannot be deleted or modified. Reading from a commit log always happens from left to right (*or old to new*).



Kafka as a write-ahead log

Kafka stores all of its messages on disk. Since all reads and writes happen in sequence, Kafka takes advantage of sequential disk reads (*more on this later*).

Kafka use cases

Kafka can be used for collecting big data and real-time analysis. Here are some of its top use cases:

1. **Metrics**: Kafka can be used to collect and aggregate monitoring data. Distributed services can push different operational metrics to Kafka servers. These metrics can then be pulled from Kafka to produce aggregated statistics.
2. **Log Aggregation**: Kafka can be used to collect logs from multiple sources and make them available in a standard format to multiple consumers.
3. **Stream processing**: Kafka is quite useful for use cases where the collected data undergoes processing at multiple stages. For example, the raw data consumed from a topic is transformed, enriched, or aggregated and pushed to a new topic for further consumption. This way of data processing is known as stream processing.
4. **Commit Log**: Kafka can be used as an external commit log for any distributed system. Distributed services can log their transactions to Kafka to keep track of what is happening. This transaction data can be used for replication between nodes and also becomes very useful for disaster recovery, for example, to help failed nodes to recover their states.
5. **Website activity tracking**: One of Kafka's original use cases was to build a user activity tracking pipeline. User activities like page clicks, searches, etc., are published to Kafka into separate topics. These topics are available for subscription for a range of use cases, including real-time processing, real-time monitoring, or loading into Hadoop or data warehousing systems for offline processing and reporting.
6. **Product suggestions**: Imagine an online shopping site like amazon.com, which offers a feature of 'similar products' to suggest lookalike products that a customer could be interested in buying. To make this work, we can track every consumer action, like search queries, product clicks, time spent on any product, etc., and record these activities in Kafka. Then, a consumer application can read these messages to find correlated products that can be shown to the customer in real-time. Alternatively, since all data is persistent in Kafka, a batch job can run overnight on the 'similar product' information gathered by the system, generating an email for the customer with product suggestions.

Kafka common terms

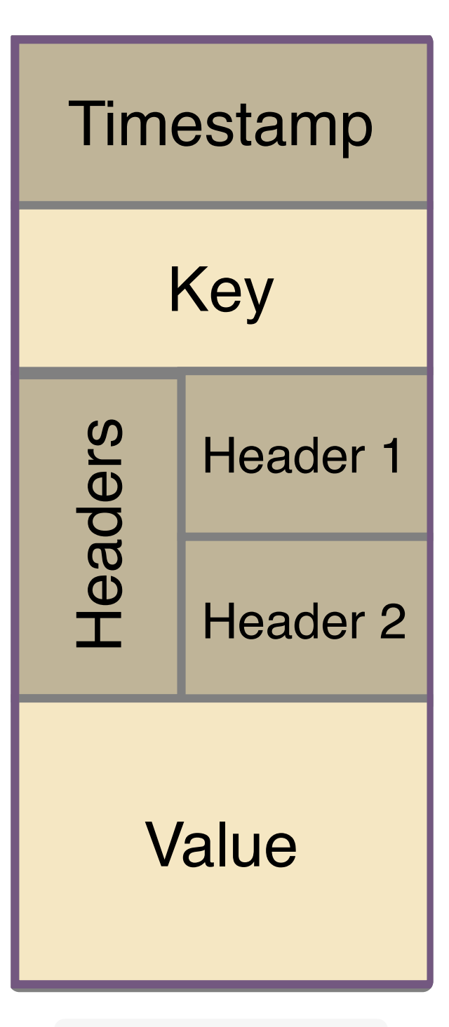
Before digging deep into Kafka's architecture, let's first go through some of its common terms.

Brokers

A Kafka server is also called a broker. Brokers are responsible for reliably storing data provided by the producers and making it available to the consumers.

Records

A record is a message or an event that gets stored in Kafka. Essentially, it is the data that travels from producer to consumer through Kafka. A record contains a key, a value, a timestamp, and optional metadata headers.



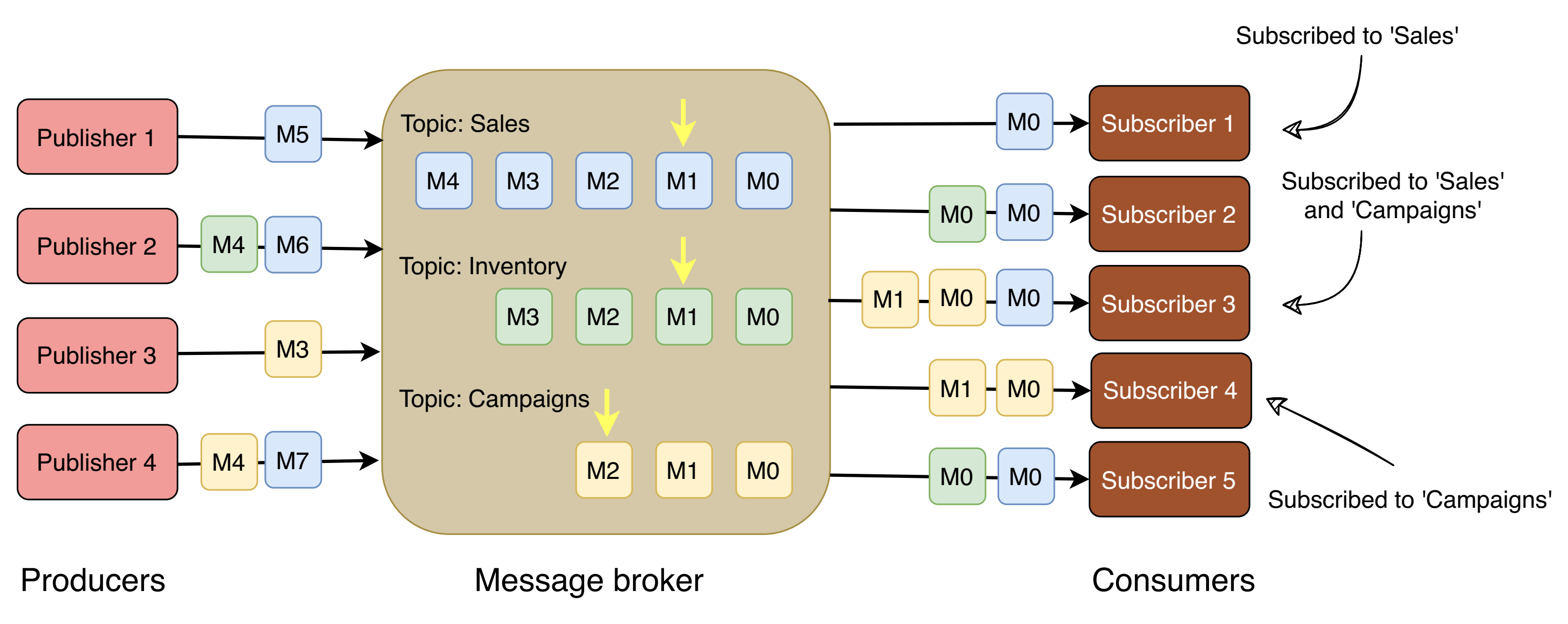
Kafka message

Topics

Kafka divides its messages into categories called Topics. In simple terms, a topic is like a table in a database, and the messages are the rows in that table.

- Each message that Kafka receives from a producer is associated with a topic.
- Consumers can subscribe to a topic to get notified when new messages are added to that topic.
- A topic can have multiple subscribers that read messages from it.
- In a Kafka cluster, a topic is identified by its name and must be unique.

Messages in a topic can be read as often as needed — unlike traditional messaging systems, messages are not deleted after consumption. Instead, Kafka retains messages for a configurable amount of time or until a storage size is exceeded. Kafka's performance is effectively constant with respect to data size, so storing data for a long time is perfectly fine.



Kafka topics

Producers

Producers are applications that publish (or write) records to Kafka.

Consumers

Consumers are the applications that subscribe to (read and process) data from Kafka topics. Consumers subscribe to one or more topics and consume published messages by pulling data from the brokers.

In Kafka, producers and consumers are fully decoupled and agnostic of each other, which is a key design element to achieve the high scalability that Kafka is known for. For example, producers never need to wait for consumers.

High-level architecture

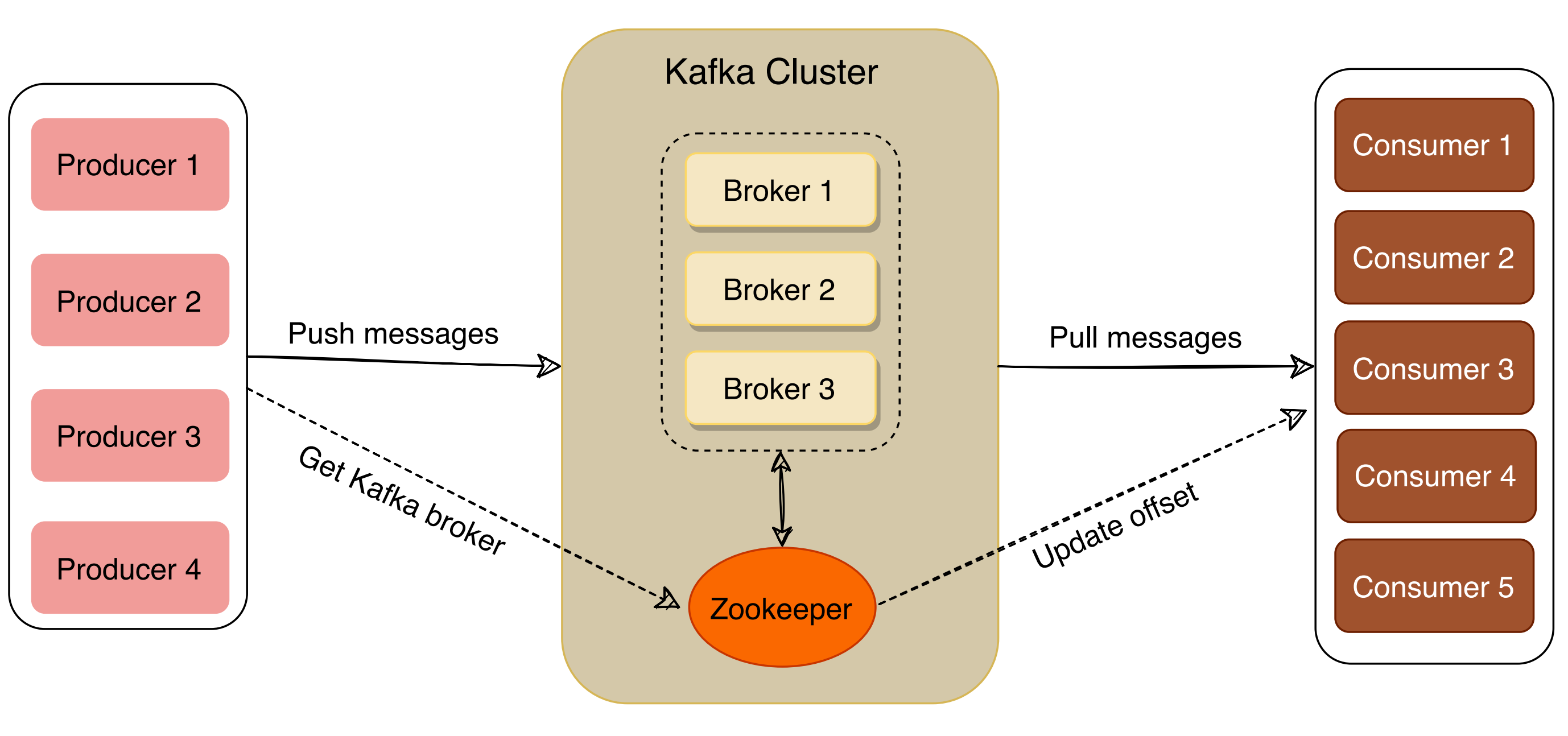
At a high level, applications (producers) send messages to a Kafka broker, and these messages are read by other applications called consumers. Messages get stored in a topic, and consumers subscribe to the topic to receive new messages.

Kafka cluster

Kafka is deployed as a cluster of one or more servers, where each server is responsible for running one Kafka broker.

ZooKeeper

ZooKeeper is a distributed key-value store and is used for coordination and storing configurations. It is highly optimized for reads. Kafka uses ZooKeeper to coordinate between Kafka brokers; ZooKeeper maintains metadata information about the Kafka cluster. We will be looking into this in detail later.



High level architecture of Kafka

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